Ahmed Yassine Hachem

Amberg, Germany | ahmedhachem0420@gmail.com | Portfolio

Education

Ostbayerische Technische Hochschule Amberg-Weiden — Amberg, Germany
Bachelor in Artificial Intelligence (Exchange Program) — *March 2026 – Present*

Pristini School of AI — Sousse, Tunisia
Bachelor in Applied Artificial Intelligence — *Sep 2023 – Jan 2026*

Experience

AI Engineer Intern — April 2026 – Present
Automotive Team, OTH-AW — *Amberg, Germany*

- Fused LiDAR, IMU and wheel odometry to produce real-time point clouds and 3D meshes of tunnels
- Adapted AI models for edge devices and preprocessed real-world data into formats they accept
- Measured the sim-to-real gap and fine-tuned the models on real data so they hold up in deployment

AI Engineer — June 2025 – July 2025
Brokins FR — *France (Remote)*

- Built two production AI tools from scratch with a co-worker for Brokins, a French insurance company
- Developed a RAG chatbot that answers questions using data from many French insurance companies
- Built an insurance comparator that uses AI matching to recommend an insurer from user preferences
- Handled deployment of both tools on Brokins servers over SSH, using PuTTY and FileZilla

AI Data Labeling — June 2024 – July 2024
PURA Solutions

- Led labeling of 20,000+ product images with metadata and designed a pipeline that cut project time by 67%
- Hit 98% accuracy on 10,000+ images and reviewed 20,000 labels with the CTO, raising model accuracy by 20%

Projects

NexusBi | *React, TypeScript, Flask, REST, Gemini API*

- Made a full-stack BI app that cleans tabular data automatically (IQR outliers, imputation, encoding, scaling)
- Chained two Gemini calls: one finds feature relationships, the second writes a report grounded in them
- Isolated sessions per tenant and cached responses by file hash on a stateless, cross-origin deployment

Queue Intelligence System | *YOLOv8, BoT-SORT, CUDA, Streamlit, Gemini*

- Built a real-time multi-object tracker (YOLOv8 + BoT-SORT, CUDA/FP16), cutting ID fragmentation ∼90%
- Sent ambiguous tracks to a Gemini vision agent, which returns a JSON verdict in a fixed schema
- Released a Streamlit app with zone annotation and a live analytics dashboard as a public demo

FertiFy | *Python, Vision-LLM, Embeddings, Gemini API*

- Designed a two-stage multimodal RAG where a vision LLM describes symptoms in an image for similarity search
- Put retrieval behind a common interface so storage can change without touching embedding or generation
- Created prompt schemas that hold Gemini to fixed output formats and branch on its confidence

Technical Skills

Programming Languages: Python, C, SQL, Java
Robotics & Simulation: ROS 2, LiDAR-Inertial SLAM (FAST-LIO2), PCL, KISS-Matcher
ML/AI: TensorFlow, Keras, scikit-learn, RAG, Multi-Agent Systems, Pandas, NumPy, MLOps, MLflow
Tools: VS Code, Eclipse IDE, AWS, Git, Power BI
Languages: English (C1), French (B2)